CS 148a PS2

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§1 Dropout Equivalence to ℓ_2 Regularization

Let $X \in \mathbb{R}^{n\times d}$ which has columns $x_1, \ldots, x_d$. Dropout multiplies each feature, call it j, by an independent mask variable $r_j \sim \text{Bernoulli}(1-p)$:

$$\tilde{X} = X \cdot \text{diag}(r_1, \ldots, r_d)$$

The dropout loss is $L(w) = \|y - \tilde{X}w\|^2$. Expanding the expectation:

$$\mathbb{E}_r\Big[\|y - \tilde{X}w\|^2\Big] = \|y\|^2 - 2\,y^\top \mathbb{E}[\tilde{X}w] + \mathbb{E}[w^\top \tilde{X}^\top \tilde{X}\, w]$$

$$\mathbb{E}[r_j] = 1 - p \Rightarrow \mathbb{E}[\tilde{X}w] = (1-p)\,Xw$$

For the quaratic term, $w^\top \tilde{X}^\top \tilde{X}\, w = \sum_{i,j} r_i\, r_j\, w_i\, w_j\, (X^\top X)_{ij}$ and use the Bernoulli moments:

$$\mathbb{E}[r_i\, r_j] = \begin{cases} (1-p)^2 & i \neq j \\ (1-p) & i = j \end{cases}$$

Diagonal:

$$\mathbb{E}[w^\top \tilde{X}^\top \tilde{X}\, w] = (1-p)^2 \sum_{i,j} w_i w_j (X^\top X)_{ij} + \big[(1-p) - (1-p)^2\big] \sum_i w_i^2 (X^\top X)_{ii}$$

$$= (1-p)^2\, w^\top X^\top X\, w + p(1-p) \sum_i w_i^2 (X^\top X)_{ii}.$$

Diagonal matrix of feature norms (from the Gram matrix):

$$\Gamma = \text{diag}\left(\sqrt{(X^\top X)_{11}}, \ldots, \sqrt{(X^\top X)_{dd}}\right).$$

where

$$\sum_i w_i^2 (X^\top X)_{ii} = \|\Gamma w\|^2.$$

Combining:

$$\mathbb{E}[L(w)] = \|y\|^2 - 2(1-p)\, y^\top Xw + (1-p)^2\, w^\top X^\top Xw + p(1-p)\, \|\Gamma w\|^2$$

$$= \boxed{\|y - (1-p)\, Xw\|^2 + p(1-p)\, \|\Gamma w\|^2}$$

§2 Batch Normalization Backpropagation

$$\mu_j = \frac{1}{N}\sum_{i=1}^{N} X_{ij}$$

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N} (X_{ij} - \mu_j)^2$$

$$\hat{X}_{ij} = \frac{X_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}}$$

$$Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$$

§2.1 $\frac{\partial L}{\partial \beta}$

Since $Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$, the parameter β_j contributes additively to every sample i in column j, so $\frac{\partial Y_{ij}}{\partial \beta_j} = 1$.

$$\boxed{\frac{\partial L}{\partial \beta_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \frac{\partial Y_{ij}}{\partial \beta_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}}}$$

§2.2 $\frac{\partial L}{\partial \gamma}$

From $Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$, $\frac{\partial Y_{ij}}{\partial \gamma_j} = \hat{X}_{ij}$.

$$\boxed{\frac{\partial L}{\partial \gamma_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \frac{\partial Y_{ij}}{\partial \gamma_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \hat{X}_{ij}}$$

§2.3 $\frac{\partial L}{\partial X}$

$$\frac{\partial L}{\partial X_{ij}} = \underbrace{\frac{\partial L}{\partial \hat{X}_{ij}} \cdot \frac{\partial \hat{X}_{ij}}{\partial X_{ij}}}_{\text{Path 1: direct}} + \underbrace{\frac{\partial L}{\partial \mu_j} \cdot \frac{\partial \mu_j}{\partial X_{ij}}}_{\text{Path 2: through } \mu_j} + \underbrace{\frac{\partial L}{\partial \sigma_j^2} \cdot \frac{\partial \sigma_j^2}{\partial X_{ij}}}_{\text{Path 3: through } \sigma_j^2}$$

$$\frac{\partial L}{\partial \hat{X}_{ij}} = \frac{\partial L}{\partial Y_{ij}} \cdot \gamma_j$$

Through $\hat{X}_{ij}$:

$$\frac{\partial \hat{X}_{ij}}{\partial X_{ij}} = \frac{1}{\sqrt{\sigma_j^2 + \varepsilon}}$$

Through μ_j:

$$\frac{\partial \mu_j}{\partial X_{ij}} = \frac{1}{N}$$

For $\frac{\partial L}{\partial \mu_j}$ (given that μ_j affects every $\hat{X}_{kj}$ in the batch):

$$\frac{\partial L}{\partial \mu_j} = \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot \frac{\partial \hat{X}_{kj}}{\partial \mu_j} = \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot \frac{-1}{\sqrt{\sigma_j^2 + \varepsilon}} = \frac{-1}{\sqrt{\sigma_j^2 + \varepsilon}} \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}}$$

Through σ_j^2:

$$\frac{\partial \sigma_j^2}{\partial X_{ij}} = \frac{2(X_{ij} - \mu_j)}{N}$$

For $\frac{\partial L}{\partial \sigma_j^2}$, since σ_j^2 affects every $\hat{X}_{kj}$ in the batch through the denominator $(\sigma_j^2 + \varepsilon)^{-1/2}$:

$$\frac{\partial L}{\partial \sigma_j^2} = \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot \frac{\partial \hat{X}_{kj}}{\partial \sigma_j^2} = \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot (X_{kj} - \mu_j) \cdot \left(-\frac{1}{2}\right) (\sigma_j^2 + \varepsilon)^{-3/2}$$

$$= -\frac{1}{2}(\sigma_j^2 + \varepsilon)^{-3/2} \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot (X_{kj} - \mu_j)$$

Combining

$$\boxed{\frac{\partial L}{\partial X_{ij}} = \frac{\partial L}{\partial \hat{X}_{ij}} \cdot \frac{1}{\sqrt{\sigma_j^2 + \varepsilon}} + \frac{\partial L}{\partial \mu_j} \cdot \frac{1}{N} + \frac{\partial L}{\partial \sigma_j^2} \cdot \frac{2(X_{ij} - \mu_j)}{N}}$$

where:

$$\frac{\partial L}{\partial \hat{X}_{ij}} = \frac{\partial L}{\partial Y_{ij}} \cdot \gamma_j$$

$$\frac{\partial L}{\partial \mu_j} = \frac{-1}{\sqrt{\sigma_j^2 + \varepsilon}} \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}}$$

$$\frac{\partial L}{\partial \sigma_j^2} = -\frac{1}{2}(\sigma_j^2 + \varepsilon)^{-3/2} \sum_{k=1}^{N} \frac{\partial L}{\partial \hat{X}_{kj}} \cdot (X_{kj} - \mu_j)$$

§3 Layer Normalization Backpropagation

§3.1 Gradient Calculations

$$\mu_i = \frac{1}{D}\sum_{j=1}^{D} X_{ij}$$

$$\sigma_i^2 = \frac{1}{D}\sum_{j=1}^{D} (X_{ij} - \mu_i)^2$$

$$\hat{X}_{ij} = \frac{X_{ij} - \mu_i}{\sqrt{\sigma_i^2 + \varepsilon}}$$

$$Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$$

§3.1.1 $\frac{\partial L}{\partial \beta}$

Since $Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$, the parameter β_j contributes additively to every sample i in column j, so $\frac{\partial Y_{ij}}{\partial \beta_j} = 1$.

$$\boxed{\frac{\partial L}{\partial \beta_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \frac{\partial Y_{ij}}{\partial \beta_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}}}$$

§3.1.2 $\frac{\partial L}{\partial \gamma}$

From $Y_{ij} = \gamma_j \hat{X}_{ij} + \beta_j$, $\frac{\partial Y_{ij}}{\partial \gamma_j} = \hat{X}_{ij}$.

$$\boxed{\frac{\partial L}{\partial \gamma_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \frac{\partial Y_{ij}}{\partial \gamma_j} = \sum_{i=1}^{N} \frac{\partial L}{\partial Y_{ij}} \cdot \hat{X}_{ij}}$$

§3.1.3 $\frac{\partial L}{\partial X}$

$$\frac{\partial L}{\partial X_{ij}} = \underbrace{\frac{\partial L}{\partial \hat{X}_{ij}} \cdot \frac{\partial \hat{X}_{ij}}{\partial X_{ij}}}_{\text{Path 1: direct}} + \underbrace{\frac{\partial L}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial X_{ij}}}_{\text{Path 2: through } \mu_i} + \underbrace{\frac{\partial L}{\partial \sigma_i^2} \cdot \frac{\partial \sigma_i^2}{\partial X_{ij}}}_{\text{Path 3: through } \sigma_i^2}$$

$$\frac{\partial L}{\partial \hat{X}_{ij}} = \frac{\partial L}{\partial Y_{ij}} \cdot \gamma_j$$

Through $\hat{X}_{ij}$:

$$\frac{\partial \hat{X}_{ij}}{\partial X_{ij}} = \frac{1}{\sqrt{\sigma_i^2 + \varepsilon}}$$

Through μ_i:

$$\frac{\partial \mu_i}{\partial X_{ij}} = \frac{1}{D}$$

For $\frac{\partial L}{\partial \mu_i}$ (given that μ_i affects every $\hat{X}_{ik}$ across the feature dimension):

$$\frac{\partial L}{\partial \mu_i} = \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot \frac{\partial \hat{X}_{ik}}{\partial \mu_i} = \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot \frac{-1}{\sqrt{\sigma_i^2 + \varepsilon}} = \frac{-1}{\sqrt{\sigma_i^2 + \varepsilon}} \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}}$$

Through σ_i^2:

$$\frac{\partial \sigma_i^2}{\partial X_{ij}} = \frac{2(X_{ij} - \mu_i)}{D}$$

For $\frac{\partial L}{\partial \sigma_i^2}$, since σ_i^2 affects every $\hat{X}_{ik}$ across the feature dimension through the denominator $(\sigma_i^2 + \varepsilon)^{-1/2}$:

$$\frac{\partial L}{\partial \sigma_i^2} = \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot \frac{\partial \hat{X}_{ik}}{\partial \sigma_i^2} = \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot (X_{ik} - \mu_i) \cdot \left(-\frac{1}{2}\right) (\sigma_i^2 + \varepsilon)^{-3/2}$$

$$= -\frac{1}{2}(\sigma_i^2 + \varepsilon)^{-3/2} \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot (X_{ik} - \mu_i)$$

Combining

$$\boxed{\frac{\partial L}{\partial X_{ij}} = \frac{\partial L}{\partial \hat{X}_{ij}} \cdot \frac{1}{\sqrt{\sigma_i^2 + \varepsilon}} + \frac{\partial L}{\partial \mu_i} \cdot \frac{1}{D} + \frac{\partial L}{\partial \sigma_i^2} \cdot \frac{2(X_{ij} - \mu_i)}{D}}$$

where:

$$\frac{\partial L}{\partial \hat{X}_{ij}} = \frac{\partial L}{\partial Y_{ij}} \cdot \gamma_j$$

$$\frac{\partial L}{\partial \mu_i} = \frac{-1}{\sqrt{\sigma_i^2 + \varepsilon}} \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}}$$

$$\frac{\partial L}{\partial \sigma_i^2} = -\frac{1}{2}(\sigma_i^2 + \varepsilon)^{-3/2} \sum_{k=1}^{D} \frac{\partial L}{\partial \hat{X}_{ik}} \cdot (X_{ik} - \mu_i)$$

§3.2 BatchNorm LayerNorm Puzzle

Let X be symmetric 16×16 matrix with $\gamma = \mathbf{1}$ and $\beta = \mathbf{0}$:

BatchNorm normalizes each column j across the batch dimension i:

$$Y_{ij} = \frac{X_{ij} - \mu_j^{\text{BN}}}{\sqrt{(\sigma_j^{\text{BN}})^2 + \varepsilon}}, \qquad \mu_j^{\text{BN}} = \frac{1}{16} \sum_{i=1}^{16} X_{ij}, \qquad (\sigma_j^{\text{BN}})^2 = \frac{1}{16} \sum_{i=1}^{16} (X_{ij} - \mu_j^{\text{BN}})^2$$

LayerNorm normalizes each row i across the feature dimension j:

$$Z_{ij} = \frac{X_{ij} - \mu_i^{\text{LN}}}{\sqrt{(\sigma_i^{\text{LN}})^2 + \varepsilon}}, \qquad \mu_i^{\text{LN}} = \frac{1}{16} \sum_{j=1}^{16} X_{ij}, \qquad (\sigma_i^{\text{LN}})^2 = \frac{1}{16} \sum_{j=1}^{16} (X_{ij} - \mu_i^{\text{LN}})^2$$

Since X is symmetric, column j of X equals row j of X. So the mean and variance of column j equal the mean and variance of row j:

$$\mu_j^{\text{BN}} = \frac{1}{16} \sum_{i=1}^{16} X_{ij} = \frac{1}{16} \sum_{i=1}^{16} X_{ji} = \mu_j^{\text{LN}}$$

$$(\sigma_j^{\text{BN}})^2 = \frac{1}{16} \sum_{i=1}^{16} (X_{ij} - \mu_j^{\text{BN}})^2 = (\sigma_j^{\text{LN}})^2$$

We can noe express Z_{ij} and using $X_{ij} = X_{ji}$ together w/ $\mu_i^{\text{LN}} = \mu_i^{\text{BN}}$ and $(\sigma_i^{\text{LN}})^2 = (\sigma_i^{\text{BN}})^2$:

$$Z_{ij} = \frac{X_{ij} - \mu_i^{\text{LN}}}{\sqrt{(\sigma_i^{\text{LN}})^2 + \varepsilon}} = \frac{X_{ji} - \mu_i^{\text{BN}}}{\sqrt{(\sigma_i^{\text{BN}})^2 + \varepsilon}} = Y_{ji}$$

So $Z = Y^\top$. Since Bob knows all of Y, he can compute every entry of Z!

$$\boxed{Z = Y^\top, \text{so Bob knows all 256 elements of } Z.}$$

What if γ and β are arbitrary?

With general γ_j and β_j, the transformation is applied per feature (col index j) in both BatchNorm and LayerNorm:

$$Y_{ij} = \gamma_j \cdot \frac{X_{ij} - \mu_j^{\text{BN}}}{\sqrt{(\sigma_j^{\text{BN}})^2 + \varepsilon}} + \beta_j$$

$$Z_{ij} = \gamma_j \cdot \frac{X_{ij} - \mu_i^{\text{LN}}}{\sqrt{(\sigma_i^{\text{LN}})^2 + \varepsilon}} + \beta_j$$

From Y, Bob can recover the normalized values $\hat{X}_{ij}^{\text{BN}} = (Y_{ij} - \beta_j)/\gamma_j$ (assuming $\gamma_j \neq 0$). By symmetry, $\hat{X}_{ij}^{\text{LN}} = \hat{X}_{ji}^{\text{BN}}$, so:

$$Z_{ij} = \gamma_j \cdot \hat{X}_{ji}^{\text{BN}} + \beta_j = \gamma_j \cdot \frac{Y_{ji} - \beta_i}{\gamma_i} + \beta_j$$

Bob can compute this for every (i, j) since he knows Y, γ, and β.

$$\boxed{\text{Bob still knows all 256 elements.}}$$

§4 Regularization, Optimizers, Augmentation in MNIST

§4.1 4B: SGD and Weight Decay

Let the objective be $L(\theta)$ and now lets take into account ℓ_2 regularization on the weights:

$$\widetilde{L}(\theta) \;=\; L(\theta) + \frac{\lambda}{2}\|\theta\|_2^2.$$

Then

$$\nabla_\theta \widetilde{L}(\theta) \;=\; \nabla_\theta L(\theta) + \lambda\theta.$$

SGD update with stepsize α is

$$\theta_{t+1} \;=\; \theta_t - \alpha\nabla_\theta \widetilde{L}(\theta_t) \;=\; \theta_t - \alpha\big(\nabla_\theta L(\theta_t) + \lambda\theta_t\big).$$

$$\theta_{t+1} \;=\; (1-\alpha\lambda)\,\theta_t \;-\; \alpha\,\nabla_\theta L(\theta_t).$$

Weight decay for SGD is defined as shrinking the weights by a multiplicative factor each step, then applying the gradient step:

$$\theta_{t+1} \;=\; (1-\alpha\lambda)\,\theta_t \;-\; \alpha\,\nabla_\theta L(\theta_t).$$

These are identical.

§4.2 4C: Optimizers

§4.2.1 Derive the Adam bias correction term $\frac{1}{1-\beta^t}$

Adam has EMAs as follows:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t, \qquad m_0 = 0,$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2, \qquad v_0 = 0,$$

where $g_t = \nabla_\theta f_t(\theta_{t-1})$.

Let's first undo the recursion for m_t:

$$m_t = (1-\beta_1)\sum_{k=1}^{t}\beta_1^{t-k}g_k.$$

$$\mathbb{E}[m_t] = (1-\beta_1)\sum_{k=1}^{t}\beta_1^{t-k}\mu = (1-\beta_1)\mu\sum_{r=0}^{t-1}\beta_1^{r} = (1-\beta_1)\mu\cdot\frac{1-\beta_1^t}{1-\beta_1} = (1-\beta_1^t)\mu.$$

So m_t is biased toward 0 (because $m_0 = 0$).

$$\widetilde{m}_t \;=\; \frac{m_t}{1-\beta_1^t}.$$

Then

$$\mathbb{E}[\widetilde{m}_t] \;=\; \frac{\mathbb{E}[m_t]}{1-\beta_1^t} \;=\; \mu,$$

so dividing by $(1-\beta_1^t)$ removes the bias.

Next, undoing v_t gives

$$v_t = (1-\beta_2)\sum_{k=1}^{t}\beta_2^{t-k}g_k^2.$$

Assuming $\mathbb{E}[g_k^2] = \nu$:

$$\mathbb{E}[v_t] = (1-\beta_2)\nu \sum_{r=0}^{t-1} \beta_2^r = (1-\beta_2^t)\nu.$$

Let

$$\widetilde{v}_t \;=\; \frac{v_t}{1-\beta_2^t} \quad \Rightarrow \quad \mathbb{E}[\widetilde{v}_t] = \nu.$$

This means the bias correction factors $\frac{1}{1-\beta_1^t}$ and $\frac{1}{1-\beta_2^t}$ are necessary to unbias.

§4.2.2 Why adding an ℓ_2 penalty to the loss is problematic for momentum/RMSProp/Adam

¶ SGD For SGD, adding $\frac{\lambda}{2}\|\theta\|^2$ results in the normal weight decay:

$$\theta_{t+1} = (1-\alpha\lambda)\theta_t - \alpha g_t,$$

Now with momentum. We can use

$$u_t = \mu u_{t-1} + g_t, \qquad \theta_{t+1} = \theta_t - \alpha u_t.$$

If we add ℓ_2 to the loss, the gradient becomes $g_t + \lambda\theta_t$:

$$u_t = \mu u_{t-1} + g_t + \lambda\theta_t.$$

Now the regularization term $\lambda\theta_t$ is accumulated into the velocity and continues to influence future updates via μu_{t-1}. This not $\theta_t \mapsto (1-\alpha\lambda)\theta_t$ applied independently each step.

¶ RMSProp RMSProp scales the gradient by coordinates:

$$\theta_{t+1} = \theta_t - \alpha\frac{g_t}{\sqrt{s_t}+\varepsilon}, \qquad s_t = \rho s_{t-1} + (1-\rho)g_t^2.$$

With ℓ_2 the update uses $(g_t + \lambda\theta_t)$:

$$\theta_{t+1} = \theta_t - \alpha\frac{g_t + \lambda\theta_t}{\sqrt{s_t}+\varepsilon}.$$

Therefore the effective decay term is

$$-\alpha\frac{\lambda\theta_t}{\sqrt{s_t}+\varepsilon},$$

which depends on the adaptive denominator $\sqrt{s_t}+\varepsilon$.

¶ Adam Adam is similar:

$$\theta_{t+1} = \theta_t - \alpha\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t}+\varepsilon}.$$

If ℓ_2 is added to the loss, then g_t is replaced by $g_t + \lambda\theta_t$ and the $\lambda\theta_t$ term is provided into both moment estimates and then divided by $\sqrt{\widehat{v}_t}+\varepsilon$ and then the regularization becomes entangled with Adam's adaptivity so not true multiplicative decay on θ.

§4.2.3 Explain how AdamW correctly penalizes weights in contrast with Adam

Adding $\frac{\lambda}{2}\|\theta\|^2$ to the loss changes the gradient to $g_t + \lambda\theta_t$, so Adam essensially applies its adaptive transform to both g_t and $\lambda\theta_t$.

AdamW applies weight decay directly to the parameters, separate from the adaptive gradient step:

$$\theta_{t+1} = \theta_t - \alpha\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t}+\varepsilon} - \alpha\lambda\theta_t = (1-\alpha\lambda)\theta_t - \alpha\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t}+\varepsilon}.$$

Hence AdamW enforces true multiplicative reduction $(1-\alpha\lambda)$ on θ independent of the adaptive scaling.